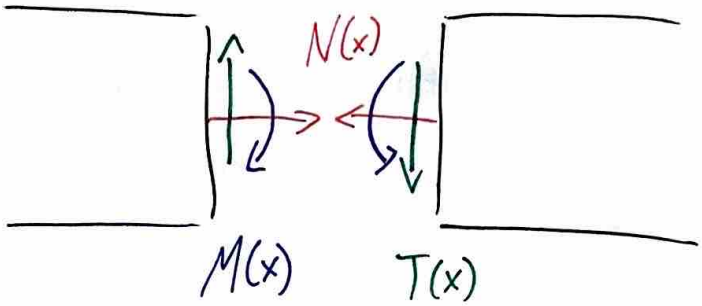
Övning 7 T- och M-diagram

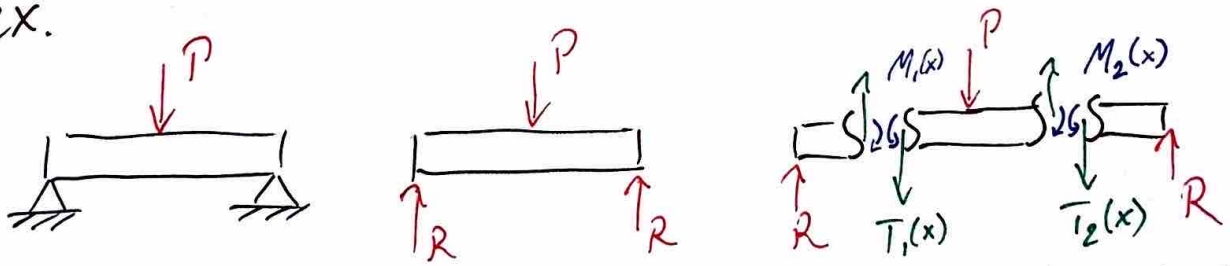
2.4: 19, 20, 28, 29

Snittkrafter

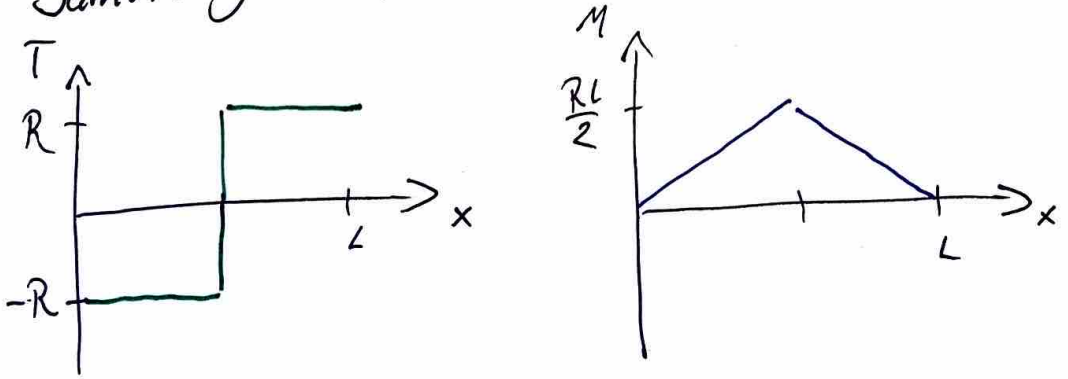


Snittkrafterna varierar ofta genom balken. Därför kan det vara bra att rita upp grafer hur de förändras för att ge förståelse av beteendet

ex.

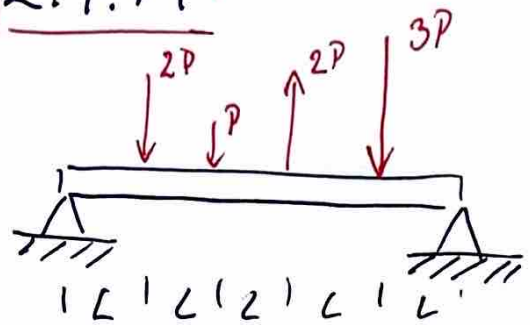


Jämvikt ger $T_1(x) = -R$ $T_2(x) = R$ $M_1(x) = Rx$ $M_2(x) = RL - Rx$

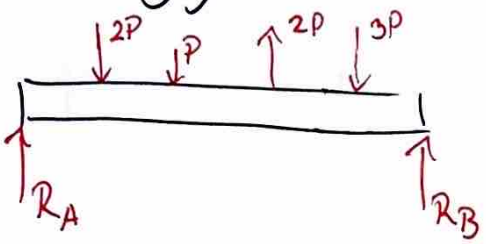


Samma skjuvkraft genom stången men max moment i mitten av stången

2.4.19



① Frilägg



Jämvikt

$\uparrow: R_A + R_B - 2P - P + 2P - 3P = 0$

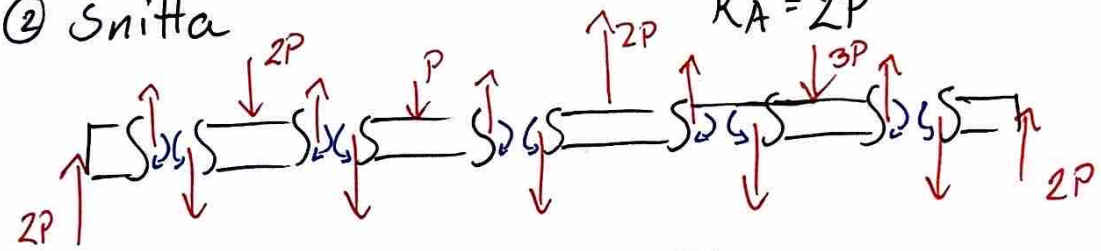
$R_A = 4P - R_B$

$\circlearrowleft: 2PL + 2PL - 6PL + 12PL - 3R_B L = 0$

$R_B = \frac{+10P}{5} = +2P$

$R_A = 2P$

② Snitta



I $\uparrow: 2P + T = 0$ $\quad \boxed{T(x) = -2P}$

$\circlearrowleft: M(x) + 2Px = 0$ $\quad \boxed{M(x) = -2Px}$

II $\uparrow: 2P - 2P + T = 0$ $\quad \boxed{T(x) = 0}$

$\circlearrowleft: M(x) + 2Px - 2P(x-L) = 0$ $\quad \boxed{M(x) = -2PL}$

III $\uparrow: 2P - 2P - P + T = 0$ $\quad \boxed{T(x) = P}$

$\circlearrowleft: M(x) + 2Px - 2P(x-L) - P(x-2L) = 0$ $\quad \boxed{M(x) = Px - 4PL}$

IV $\uparrow: 2P - 2P - P + 2P + T = 0$ $\quad \boxed{T(x) = -P}$

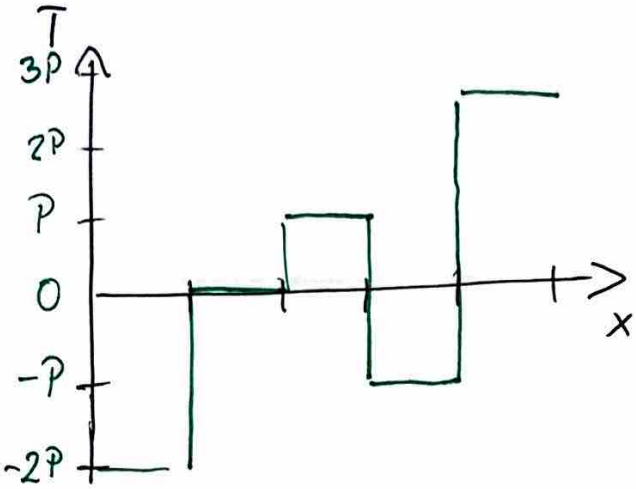
$\circlearrowleft: M(x) + 2Px - 2P(x-L) - P(x-2L) + 2P(x-3L) = 0$

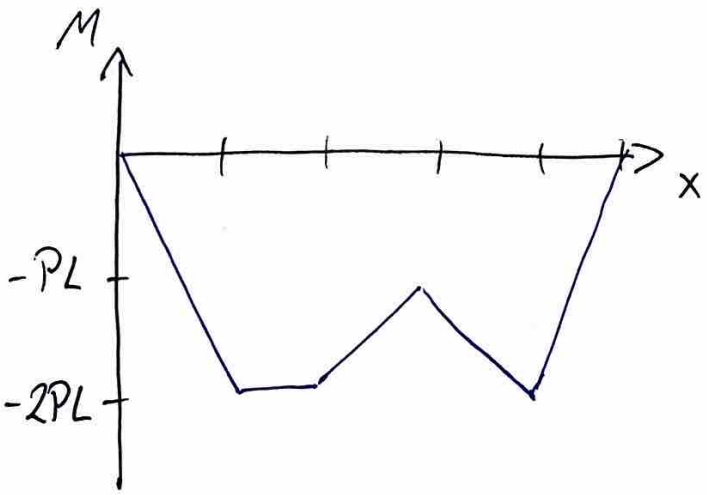
$\boxed{M(x) = -Px + 2PL}$

V $\uparrow$: $2P - 2P - P + 2P - 3P + T = 0$ $\quad T(x) = 2P$

$\circlearrowleft$: $M(x) + 2Px - 2P(x-L) - P(x-2L) + 2P(x-3L) - 3P(x-4L) = 0$

$M(x) = 2Px - 10PL$

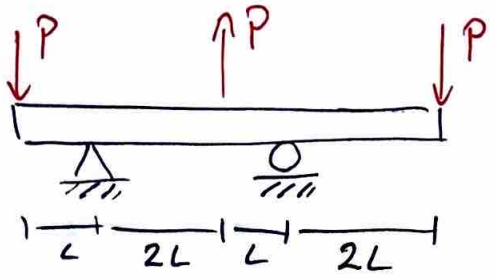




Randvillkor korrekta $M(0) = M(5L) = 0$

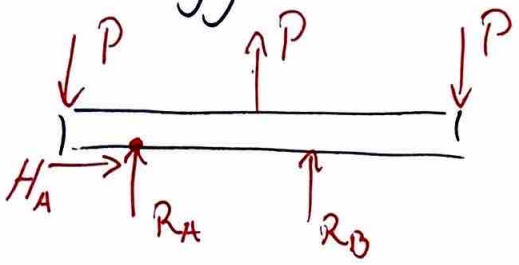
$T(x) = \frac{dM(x)}{dx}$ stämmer!

2.4.20



Sökt: T- och M-diagram

① Frilägg



Jämvikt

$\rightarrow: \; H_A = 0$

$\uparrow: \; R_A + R_B + P - P - P = 0$

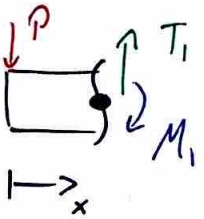
$R_A = P - R_B$

$\curvearrowright A: \; 5LP - R_B 3L - 2LP - LP = 0$

$R_B = \frac{2}{3}P$

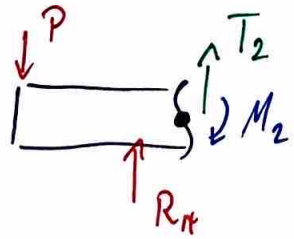
$R_A = \frac{1}{3}P$

② Snitta



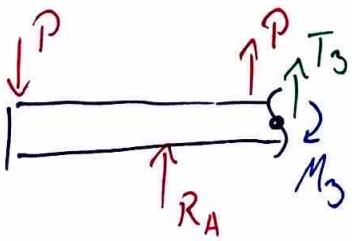
$\uparrow: \; T_1 - P = 0 \quad \Rightarrow \boxed{T_1 = P}$

$\curvearrowright \; M_1 - Px = 0 \quad \Rightarrow \boxed{M_1 = Px}$



$\uparrow: \; T_2 + R_A - P = 0 \quad \Rightarrow \boxed{T_2 = \frac{2}{3}P}$

$\curvearrowright \; M_2 + R_A(x-L) - Px \quad \Rightarrow \boxed{M_2 = \frac{2}{3}Px + \frac{1}{3}PL}$

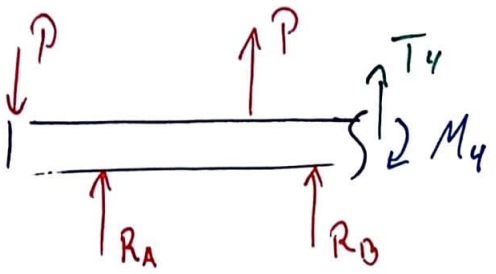


$\uparrow: \; T_3 + P + R_A - P = 0 \quad \Rightarrow \boxed{T_3 = \frac{1}{3}P}$

$\curvearrowright: \; M_3 + P(x-3L) + R_A(x-L) - Px$

$\boxed{M_3 = -\frac{1}{3}Px + \frac{10}{3}PL}$

2.4.20 del 2

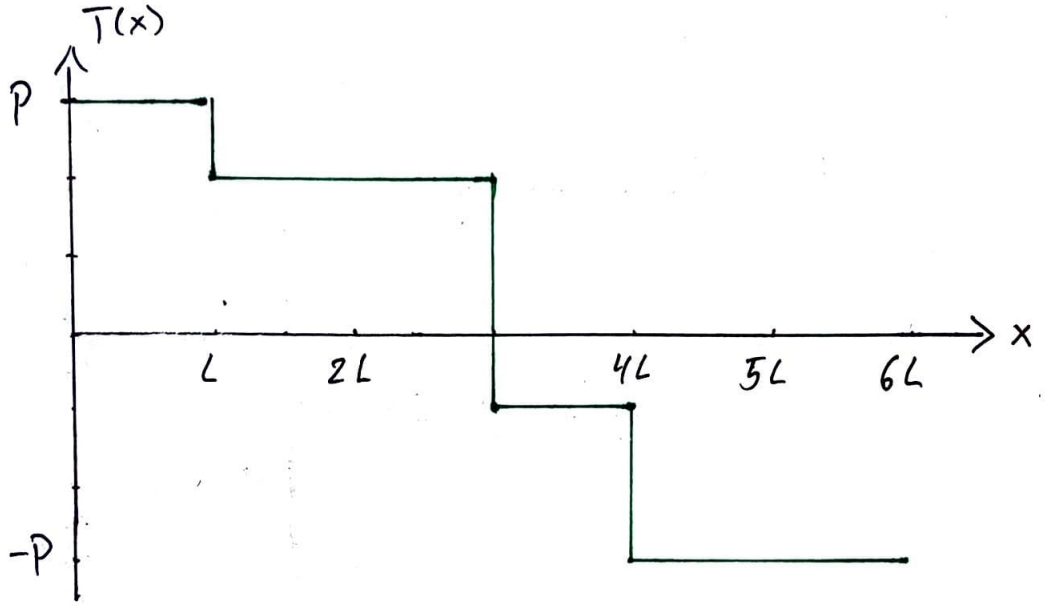


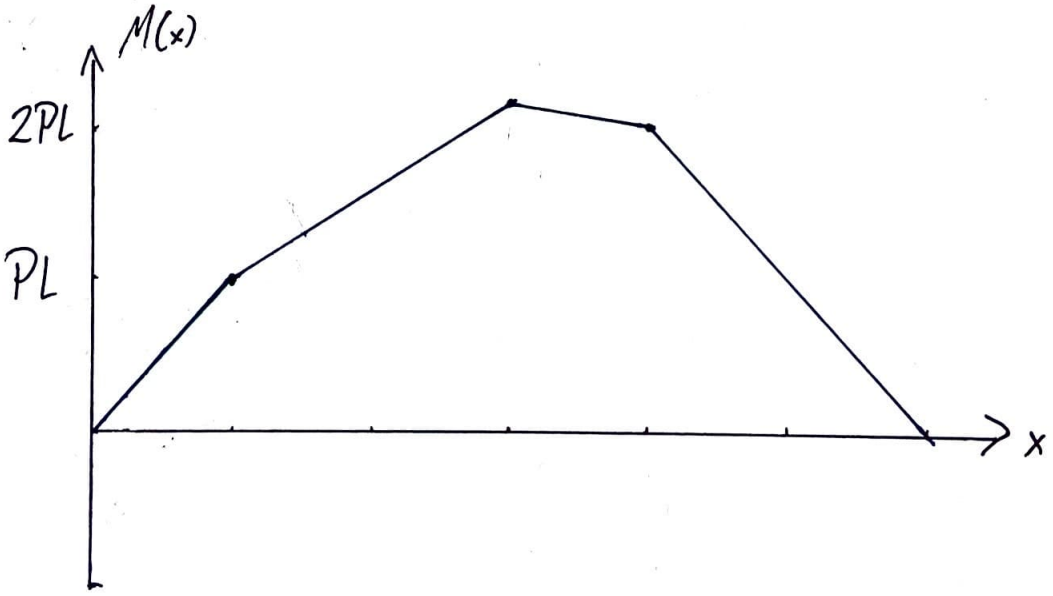
$\uparrow: T_4 + R_B + P + R_A - P = 0$

$$\boxed{T_4 = -P}$$

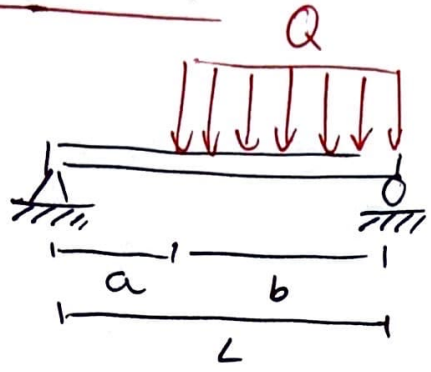
$\circlearrowright M_4 + R_B(x-4L) + P(x-3L) + R_A(x-L) - Px = 0$

$$\boxed{M_4 = -Px + 6PL}$$





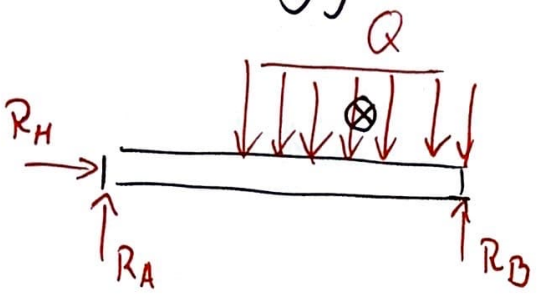
2.4:29



Sökt: T- och M-diagram

$$\boxed{a+b=L}$$

① Frilägg



$\rightarrow: \quad R_H = 0$

$\uparrow: \quad R_A - Q + R_B = 0$

$\circlearrowleft A: \quad Q(a+b/2) - R_B L = 0$

$$\boxed{R_B = \frac{Q}{L}\left(a+\frac{b}{2}\right)}$$

$$R_A = Q\left(1-\frac{a+\frac{b}{2}}{L}\right) = Q\left(1-\frac{L-\frac{b}{2}}{L}\right)$$

$$\boxed{R_A = \frac{Qb}{2L}}$$

② Snittning eller jämviktsekvationer

$q_1 = 0 \quad \Rightarrow \quad \boxed{T_1(x) = C_1} \quad \boxed{M_1(x) = C_1 x + C_2}$

Randvillkor

$T_1(0) = -R_A \quad M_1(0) = 0 \quad \Rightarrow \quad C_1 = -\frac{Qb}{2L} \quad C_2 = 0$

$q_2 = -\frac{Q}{b} \Rightarrow \boxed{T_2(x) = +\frac{Q}{b}x + C_3} \quad \boxed{M_2(x) = +\frac{Q}{2b}x^2 + C_3 x + C_4}$

Kontinuerligtet

$T_1(a) = T_2(a) \quad M_1(a) = M_2(a)$

alternativt

$T_2(L) = R_B \quad M_2(L) = 0$

2.4:29 del 2

$$T_2(L) = R_B = \frac{Q}{L}\left(a+\frac{b}{2}\right)$$

$$\frac{Q}{b}L + C_3 = \frac{Q}{L}\left(a+\frac{b}{2}\right) \qquad C_3 = \frac{Q}{L}\left(a+\frac{b}{2}\right) - \frac{Q}{b}L$$

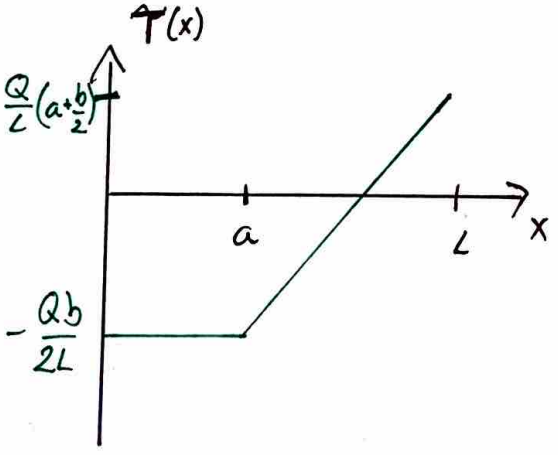
$$\boxed{T_2(x) = \frac{Q}{L}\left(a+\frac{b}{2}\right) - \frac{Q(L-x)}{b}}$$

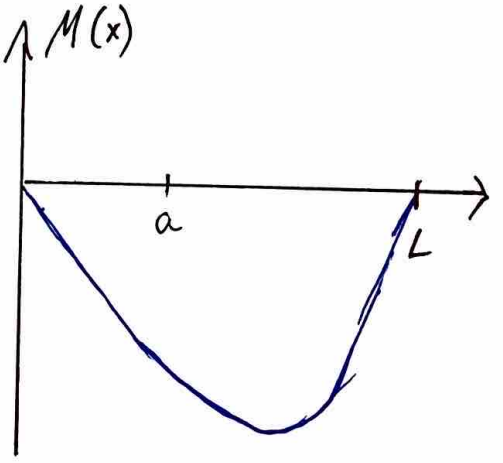
$$M_2(L) = 0$$

$$\frac{Q}{2b}L^2 + \left(\frac{Q}{L}\left(a+\frac{b}{2}\right) - \frac{Q}{b}L\right)L + C_4 = 0$$

$$\Rightarrow C_4 = \frac{QL^2}{2b} - \frac{Q}{L}\left(a+\frac{b}{2}\right)L$$

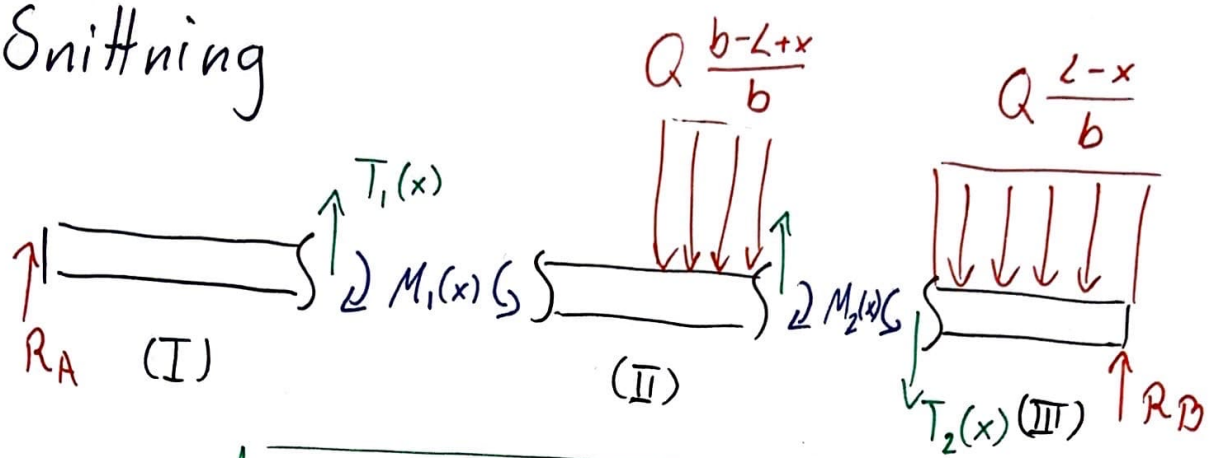
$$\boxed{M_2(x) = Q\frac{(L-x)^2}{2b} - \frac{Q}{L}\left(a+\frac{b}{2}\right)(L-x)}$$





2.4:29 alternativ lösningsväg

③ Snittning



(I) $\uparrow$: $\boxed{T_1(x) = -R_A = -\frac{Qb}{2L}}$

$\circlearrowleft$: $M_1(x) + R_A x = 0 \Rightarrow \boxed{M_1(x) = -\frac{Qb}{2L}x}$

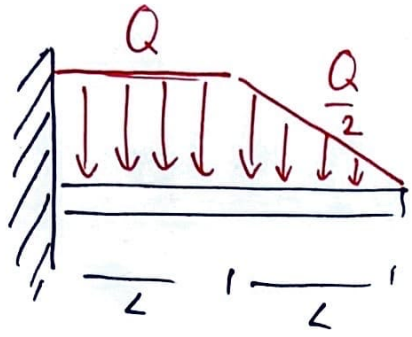
(III) $\uparrow$: $-T_2(x) - \frac{Q(L-x)}{b} = 0 \Rightarrow \boxed{T_2(x) = -\frac{Q}{b}(L-x) + R_B}$

$\circlearrowleft$: $-M_2(x) + \frac{Q(L-x)}{b}\frac{(L-x)}{2} - R_B(L-x) = 0$

$$\boxed{M_2(x) = +\frac{Q(L-x)^2}{2b} - \frac{Q}{L}\left(a+\frac{b}{2}\right)(L-x)}$$

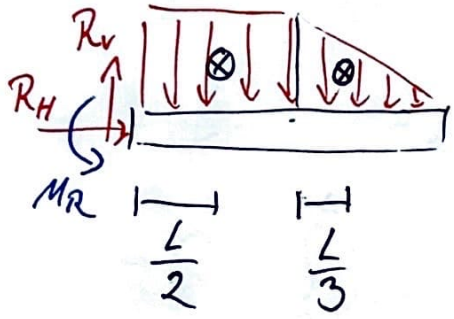
$$\boxed{\begin{aligned} T_1(x) &= -\frac{Qb}{2L} & M_1(x) &= -\frac{Qb}{2L}x \\ T_2(x) &= -\frac{Q}{b}(L-x) & M_2(x) &= \frac{Q(L-x)^2}{2b} - \frac{Q}{L}\left(a+\frac{b}{2}\right)(L-x) \end{aligned}}$$

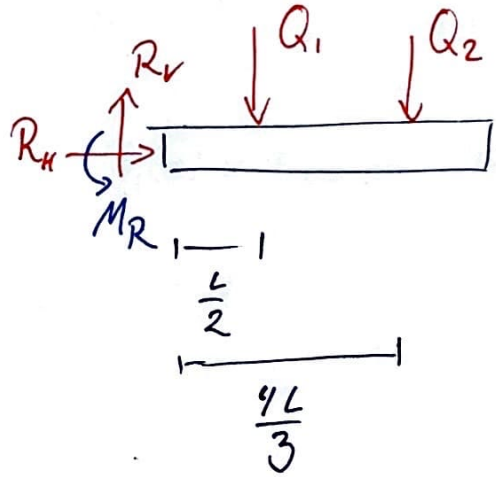
2.4:28



Sökt: T- och M-diagram

① Frilägg





② Jämvikt

$\rightarrow: R_H = 0$

$\uparrow: R_V - Q_1 - Q_2 = 0$

$\circlearrowleft: -M_R + Q_1 \frac{L}{2} + Q_2 \frac{4L}{3} = 0$

$$Q_1 = Q$$
$$Q_2 = \frac{Q}{2}$$

$\Rightarrow$ $R_V = \frac{3}{2}Q$

$M_R = \frac{7}{6}Q$

2.4:28 del 2

② Jämviktsekvationer

$$\frac{dT(x)}{dx} = -q(x) \qquad \frac{dM(x)}{dx} = T(x)$$

I vårt fall är totala lasten given

$$-Q = \int_0^L q_1(x)\,dx \qquad \text{och} \qquad -\frac{Q}{2} = \int_L^{2L} q_2(x)\,dx$$

För $0 < x < L$ $\quad q_1(x) = \text{konst} = q_1$

$$-Q = \int_0^L q_1\,dx = [q_1 x]_0^L = q_1 L \Rightarrow \boxed{q_1 = -\frac{Q}{L}}$$

För $L < x < 2L$ $\quad q_2(x) = \text{linjär} = ax + b$

$$q_2(L) = q_1 = -\frac{Q}{L} \qquad q_2(2L) = 0$$

$$\begin{cases} aL + b = -\frac{Q}{L} \\ a2L + b = 0 \end{cases} \Rightarrow \begin{cases} a = \frac{Q}{L^2} \\ b = -\frac{2Q}{L} \end{cases}$$

$$\boxed{q_2(x) = \frac{Q}{L^2}x - \frac{2Q}{L}}$$

$$\frac{dT_1(x)}{dx} = -q_1 = \frac{Q}{L} \quad \Rightarrow \boxed{T_1(x) = \frac{Q}{L}x + C_1}$$

$$\frac{dM_1(x)}{dx} = T_1(x) \quad \Rightarrow \boxed{M_1(x) = \frac{Q}{2L}x^2 + C_1 x + C_2}$$

$$\frac{dT_2(x)}{dx} = -q_2 = -\frac{Q}{L^2}x + \frac{2Q}{L} \quad \Rightarrow \boxed{T_2(x) = -\frac{Q}{2L^2}x^2 + \frac{2Q}{L}x + C_3}$$

$$\frac{dM_2(x)}{dx} = T_2(x) \quad \Rightarrow \boxed{M_2(x) = -\frac{Q}{6L^2}x^3 + \frac{2Q}{2L}x^2 + C_3 x + C_4}$$

4 konstanter som bestäms från randvillkor.

Inga punktkrafter ger kontinuerliga kurvor

$T_1(L) = T_2(L)$ (1) $\quad M_1(L) = M_2(L)$ (2)

Fri ände

$T_2(2L) = 0$ (3) $\quad M_2(2L) = 0$ (4)

Fyra villkor och fyra obekanta

$$\begin{cases} \frac{Q}{L}L + C_1 = -\frac{Q}{2L^2}L^2 + \frac{2Q}{L}L + C_3 & (1) \\ \frac{Q}{2L}L^2 + C_1 L + C_2 = -\frac{Q}{6L^2}L^3 + \frac{2Q}{2L}L^2 + C_3 L + C_4 & (2) \\ -\frac{Q}{2L^2}(2L)^2 + \frac{2Q}{L}2L + C_3 = 0 & (3) \\ -\frac{Q}{6L^2}(2L)^3 + \frac{2Q}{2L}(2L)^2 + C_3 2L + C_4 = 0 & (4) \end{cases}$$

2.4:28 del 4

$$\Rightarrow \begin{cases} C_1 = -\frac{3Q}{2} & C_2 = \frac{7}{6}QL \\ C_3 = -2Q & C_4 = \frac{4}{3}QL \end{cases}$$

$$T_1(x) = \frac{Q}{L}x - \frac{3Q}{2}$$

$$M_1(x) = \frac{Q}{2L}x^2 - \frac{3Q}{2}x + \frac{7}{6}QL \qquad 0 < x < L$$

$$T_2(x) = -\frac{Q}{2L}x^2 - \frac{2Q}{L}x - 2Q$$

$$M_2(x) = -\frac{Q}{6L}x^3 + \frac{Q}{L}x^2 - 2Qx + \frac{4}{3}QL \qquad L < x < 2L$$

